TD1 Matrice

Exo1:

[illegible] $\begin{pmatrix} [illegible] & 5 \\ [illegible] & 1 \\ [illegible] & 0 \end{pmatrix}_{3\times2}$ $N = \begin{pmatrix} 0 & -2 \\ 3 & -1 \\ 0 & 1 \end{pmatrix}_{3\times2}$ $L = \begin{pmatrix} 1 & 5 \\ -1 & 1 \\ 0 & 1 \end{pmatrix}_{3\times2}$ $K = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 1 & 3 \end{pmatrix}_{2\times3}$

[illegible] $= \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & -1 & -1 \\ 1 & 2 & -1 & -1 \\ -1 & 1 & 1 & 0 \end{pmatrix}_{4\times4}$

$M+N = \begin{pmatrix} 1 & 3 \\ 5 & 0 \\ 3 & 1 \end{pmatrix}$; $M-N = \begin{pmatrix} 1 & 7 \\ -1 & 2 \\ 3 & -1 \end{pmatrix}$; $N+3M = \begin{pmatrix} 3 & 13 \\ 9 & 2 \\ 9 & 1 \end{pmatrix}$

$3M+2N = \begin{pmatrix} 3 & 11 \\ 12 & 1 \\ 9 & 2 \end{pmatrix}$; on peut pas calculer $J+K$, $N+J$, $K+L$ et $J+L$ car ils ne sont pas de même format

Exo2:

$A = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}_{2\times2}$; $B = \begin{pmatrix} 1 & 2 \\ -2 & -4 \end{pmatrix}_{2\times2}$; $C = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}_{3\times3}$; $D = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}_{3\times3}$; $E = \begin{pmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \end{pmatrix}_{2\times3}$

$AB = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}_{2\times2}$; $BA = \begin{pmatrix} 6 & 3 \\ -12 & -6 \end{pmatrix}_{2\times2}$; $CD = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}_{3\times3}$; $DC = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \\ 1 & 0 & 1 \end{pmatrix}_{3\times3}$

$AE = \begin{pmatrix} -1 & -2 & 3 \\ -1 & -2 & 3 \end{pmatrix}_{2\times3}$; on peut pas calculer CE car le nom de colone C et different de n° de ligne de E

Exo3:

$A = \begin{pmatrix} 2 & 5 & -1 \\ 0 & 1 & 3 \\ 0 & -2 & 4 \end{pmatrix}_{3\times3}$ $B = \begin{pmatrix} 1 & 7 & -1 \\ 2 & 3 & 4 \\ 0 & 0 & 0 \end{pmatrix}_{3\times3}$ $C = \begin{pmatrix} 1 & 2 \\ 0 & 4 \\ -1 & 0 \end{pmatrix}_{3\times2}$

$(A-2B)C = \begin{pmatrix} 0 & -9 & 1 \\ -4 & -5 & -5 \\ 0 & -2 & 4 \end{pmatrix}_{3\times3} \begin{pmatrix} 1 & 2 \\ 0 & 4 \\ -1 & 0 \end{pmatrix}_{3\times2} = \begin{pmatrix} -1 & -36 \\ 1 & -28 \\ -4 & -8 \end{pmatrix}$

$C^T.A = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 4 & 0 \end{pmatrix}_{2\times3} . A = \begin{pmatrix} 2 & 7 & -5 \\ 4 & 14 & 10 \end{pmatrix}$

$C^T.B = \begin{pmatrix} 1 & 7 & -1 \\ 10 & 26 & 14 \end{pmatrix}$; $A^t = \begin{pmatrix} 2 & 0 & 0 \\ 5 & 1 & -2 \\ -1 & 3 & 4 \end{pmatrix}$; $B^t = \begin{pmatrix} 1 & 2 & 0 \\ 7 & 3 & 0 \\ -1 & 4 & 0 \end{pmatrix}$

$C^t(A^t - 2B^t) = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 4 & 0 \end{pmatrix}_{2\times3} \begin{pmatrix} 0 & -4 & 0 \\ -9 & -5 & -2 \\ 1 & -5 & 4 \end{pmatrix}_{3\times3} = \begin{pmatrix} -1 & 1 & -4 \\ -36 & -28 & -8 \end{pmatrix}$

Exo4 :

$\begin{bmatrix}-1&-2\\3&4\end{bmatrix}$; $B=\begin{bmatrix}3&1\\7&2\end{bmatrix}$; $C=\begin{bmatrix}1&3&0\\2&1&-1\\-2&1&1\end{bmatrix}$; $D=\begin{bmatrix}2&-2&1\\3&0&5\\1&1&8\end{bmatrix}$

$$A^{-1}=\frac{1}{|A|}\cdot\begin{bmatrix}4&2\\-3&-1\end{bmatrix}=\begin{bmatrix}2&1\\-3/2&-1/2\end{bmatrix}$$

$$B^{-1}=\frac{1}{|B|}\begin{bmatrix}2&-1\\-7&3\end{bmatrix}=\begin{bmatrix}-2&1\\7&-3\end{bmatrix}$$

$$C^{-1}=\frac{1}{|C|}\cdot {}^t\mathrm{com}(C) \Rightarrow \det(C)=1\begin{vmatrix}1&-1\\1&1\end{vmatrix}-3\begin{vmatrix}2&-1\\-2&1\end{vmatrix}=2$$

$$\mathrm{com}(C)=\begin{bmatrix}\begin{vmatrix}1&-1\\1&1\end{vmatrix}&-\begin{vmatrix}2&-1\\-2&1\end{vmatrix}&\begin{vmatrix}2&1\\-2&1\end{vmatrix}\\-\begin{vmatrix}3&0\\1&1\end{vmatrix}&\begin{vmatrix}1&0\\-2&1\end{vmatrix}&-\begin{vmatrix}1&3\\-2&1\end{vmatrix}\\\begin{vmatrix}3&0\\1&-1\end{vmatrix}&-\begin{vmatrix}1&0\\2&-1\end{vmatrix}&\begin{vmatrix}1&3\\2&1\end{vmatrix}\end{bmatrix}\Rightarrow \mathrm{Com}(C)=\begin{pmatrix}2&0&4\\-2&1&-7\\-3&1&-5\end{pmatrix}$$

$${}^t\mathrm{com}(C)=\begin{pmatrix}2&-2&-3\\0&1&1\\4&-7&-5\end{pmatrix}\Rightarrow C^{-1}=\begin{bmatrix}1&-1&-3/2\\0&1/2&1/2\\2&-7/2&-5/2\end{bmatrix}$$

$$D^{-1}=\frac{1}{|D|}\cdot {}^t\mathrm{com}(D)\Rightarrow \det(D)=3\begin{vmatrix}-2&1\\1&2\end{vmatrix}+5\begin{vmatrix}2&-2\\1&1\end{vmatrix}=4$$

$$\mathrm{Com}(D)=\begin{bmatrix}\begin{vmatrix}0&5\\1&2\end{vmatrix}&-\begin{vmatrix}3&5\\1&2\end{vmatrix}&\begin{vmatrix}3&0\\1&1\end{vmatrix}\\-\begin{vmatrix}-2&1\\1&2\end{vmatrix}&\begin{vmatrix}2&1\\1&2\end{vmatrix}&-\begin{vmatrix}2&-2\\1&1\end{vmatrix}\\\begin{vmatrix}-2&1\\1&2\end{vmatrix}&-\begin{vmatrix}2&1\\3&5\end{vmatrix}&\begin{vmatrix}2&-2\\3&0\end{vmatrix}\end{bmatrix}\Rightarrow \mathrm{Com}(D)=\begin{bmatrix}-5&-1&3\\5&3&-4\\-5&-7&2\end{bmatrix}$$

$$D^{-1}=\begin{bmatrix}-5/24&5/24&-5/24\\-1/24&3/24&-7/24\\3/2&-1&1/2\end{bmatrix}$$

Exo5 :

$$A=\begin{pmatrix}1&0&2\\0&-1&1\\1&-2&0\end{pmatrix}\qquad B=\begin{pmatrix}-1&-2&0\\2&3&0\\0&0&1\end{pmatrix}$$

1)

$$A^2=\begin{pmatrix}1&0&2\\0&-1&1\\1&-2&0\end{pmatrix}\begin{pmatrix}1&0&2\\0&-1&1\\1&-2&0\end{pmatrix}=\begin{pmatrix}3&-4&2\\1&-1&-1\\1&2&0\end{pmatrix}\cdot\begin{pmatrix}1&0&2\\0&-1&1\\1&-2&0\end{pmatrix}\Rightarrow A^3=\begin{pmatrix}5&0&2\\0&3&1\\1&-2&4\end{pmatrix}$$

$$A^3-A=\begin{pmatrix}4&0&0\\0&4&0\\0&0&4\end{pmatrix}\Rightarrow 4I_3\Rightarrow 4(A\cdot A^{-1})=4I_3\Rightarrow \cancel{4I_3}=A^3-A$$

$$\Rightarrow 4(A\cdot A^{-1})=A^3-A\Rightarrow 4A\cdot A^{-1}=A^3-A$$

Matrice TD2

Exo1:

$$A=\begin{bmatrix}1&3&0\\2&1&-1\\-2&1&1\end{bmatrix} ;\ B=\begin{bmatrix}1&3&-2&1\\2&7&3&2\\4&3&0&3\\-2&2&1&2\end{bmatrix} ;\ C=\begin{bmatrix}3&0&0&0&0\\2&5&0&0&0\\5&0&-2&0&0\\7&5&0&2&0\\3&2&-1&1&1\end{bmatrix}$$

$$* \det(A)=\begin{vmatrix}1&3&0&1&3\\2&1&-1&2&1\\-2&1&1&-2&1\end{vmatrix} \Rightarrow (1\times1\times1)+(3\times(-1)\times(-2))+0-((3\times2\times1)+(-1)+0)$$

$$\Rightarrow 7-5=\boxed{2}$$

$$* \det(B)=\left|\begin{array}{cccc|cc}1&3&-2&1&1&3\\2&7&3&2&2&7\\4&3&0&3&4&3\\-2&2&1&2&-2&2\end{array}\right| \Rightarrow (0+54-32)-(18+0-18)=\boxed{-86}$$

$$* \det(C)=\begin{vmatrix}3&0&0&0&0\\2&5&0&0&0\\5&0&-2&0&0\\7&5&0&2&0\\3&2&-1&1&1\end{vmatrix} \Rightarrow (3\times5\times(-2)\times2\times1)=\boxed{-60}$$

Exo2:

$$A=\begin{bmatrix}5&2\\6&4\end{bmatrix} ;\ B=\begin{bmatrix}2&4\\5&10\end{bmatrix} ,\ C=\begin{bmatrix}-1&3&0\\2&1&-1\\-2&1&1\end{bmatrix} ,\ D=\begin{bmatrix}-1&-2&0\\2&3&0\\0&0&1\end{bmatrix}$$

$$E=\begin{bmatrix}2&1&0&-4\\-3&0&-1&6\\-2&7&0&4\\1&0&3&-2\end{bmatrix} \quad F=\begin{bmatrix}2&0&0&0\\0&3&0&0\\0&0&1&0\\0&0&0&-2\end{bmatrix}$$

$$* A^{-1}=\frac{1}{|A|}\cdot\begin{bmatrix}4&-2\\-6&5\end{bmatrix} \Rightarrow \frac{1}{8}\begin{bmatrix}4&-2\\-6&5\end{bmatrix} \Rightarrow \begin{bmatrix}1/2&-1/4\\-3/4&5/8\end{bmatrix}$$

$$* B^{-1}=\frac{1}{|B|}\begin{bmatrix}10&-4\\-5&2\end{bmatrix} \Rightarrow \cancel{1/}$$ comme que $|B|=0$ donc la matrice n'a aucune inverse

$$* C^{-1}=\frac{1}{|C|}\cdot {}^t com(C) \Rightarrow Com(C)=\begin{bmatrix}\begin{vmatrix}1&-1\\1&1\end{vmatrix} & -\begin{vmatrix}2&-1\\-2&1\end{vmatrix} & \begin{vmatrix}2&1\\-2&1\end{vmatrix}\\ -\begin{vmatrix}3&0\\1&1\end{vmatrix} & \begin{vmatrix}-1&0\\-2&1\end{vmatrix} & -\begin{vmatrix}-1&3\\-2&1\end{vmatrix}\\ \begin{vmatrix}3&0\\1&-1\end{vmatrix} & -\begin{vmatrix}-1&0\\2&-1\end{vmatrix} & \begin{vmatrix}-1&3\\2&1\end{vmatrix}\end{bmatrix}$$

$$Com(C)=\begin{bmatrix}2&0&4\\-3&1&-5\\-3&-1&-7\end{bmatrix} \Rightarrow C^{-1}=\begin{bmatrix}-1&0&-2\\3/2&-1/2&5/2\\3/2&1/2&7/2\end{bmatrix} \Rightarrow \begin{bmatrix}-1&3/2&3/2\\0&-1/2&1/2\\-2&5/2&7/2\end{bmatrix}$$

Exo3:

$$\begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} \; ; \; B = \begin{bmatrix} 1 & -1 \\ -4 & -5 \end{bmatrix} \quad C = \begin{bmatrix} 3 & 0 & -1 \\ 2 & 1 & 2 \\ -1 & 0 & 3 \end{bmatrix} \quad D = \begin{bmatrix} 0 & -1 & 1 \\ -6 & -2 & 3 \\ -2 & -2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & -1 & 0 \\ 0 & 7 & 0 & 6 \\ 0 & 0 & 3 & 0 \end{bmatrix}$$

$$P_\lambda(A) = \det(A - \lambda I_2) \Rightarrow \left(\begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right) \Rightarrow \begin{bmatrix} 3-\lambda & 2 \\ 1 & 2-\lambda \end{bmatrix}$$

$$\Rightarrow (3-\lambda)(2-\lambda) - 2 \Rightarrow \lambda^2 - 5\lambda + 4 = 0 \Rightarrow \Delta = 25 - 4\times 1 \times 4 = 9$$

$$\lambda_1 = \frac{5-3}{2} = 1 \; ; \; \lambda_2 = \frac{5+3}{2} = 4$$

- La valeur propre λ_1 associé à V_1:

$$\det(A - 1I_2)V_1 = 0 \Rightarrow \left(\begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right)\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} 2a + 2b = 0 \\ a + b = 0 \end{cases} \begin{matrix} a = -b \\ b = -a \end{matrix} \Bigg\} \; V_1 \begin{pmatrix} a \\ -a \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$P_\lambda(B) = \det(B - \lambda I_2)V_1 \Rightarrow \left(\begin{bmatrix} 1 & -1 \\ -4 & -5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right)\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \left(\begin{bmatrix} 1-\lambda & 0 \\ -4 & -5-\lambda \end{bmatrix}\right)\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow (1-\lambda)(-5-\lambda) \Rightarrow \lambda^2 + 4\lambda - 5 = 0$$

$$\Rightarrow \Delta = 16 - 4\times 1 \times (-5) = 0 \Rightarrow 36 \Rightarrow \lambda_1 = \frac{-4-6}{2} = -5$$

$$\lambda_2 = \frac{-4+6}{2} = 1 \; ; \; \text{pour } \lambda_1 = -5$$

$$\left(\begin{bmatrix} 1 & -1 \\ -4 & -5 \end{bmatrix} - \begin{bmatrix} -5 & 0 \\ 0 & -5 \end{bmatrix}\right)\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 6 & -1 \\ -4 & 0 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} 6a - b = 0 \Rightarrow b = 0 \\ -4a = 0 \Rightarrow a = 0 \end{cases} \Rightarrow V_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

TD 3

① $\begin{cases} x + 4y = 6 \\ 3x - 2y = 4 \end{cases}$; $\underbrace{\begin{bmatrix} 1 & 4 \\ 3 & -2 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} 6 \\ 4 \end{bmatrix}}_{B}$

$\Rightarrow A \cdot X = B \Rightarrow A \cdot A^{-1} \cdot X = A^{-1} \cdot B \Rightarrow X = A^{-1} \cdot B$

$\det(A) = -14 \neq 0$

$A^{-1} = -\frac{1}{14} \begin{bmatrix} -2 & -4 \\ -3 & 1 \end{bmatrix} \Rightarrow A^{-1} \cdot B = -\frac{1}{14} \begin{bmatrix} -2 & -4 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$

$\Rightarrow X = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

② $\begin{cases} 5x + 3y = 1 \\ 2x + 2y = -2 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 5 & 3 \\ 2 & 2 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} 1 \\ -2 \end{bmatrix}}_{B}$

$\det(A) = 4 \neq 0$

$A^{-1} = \frac{1}{4} \begin{bmatrix} 2 & -3 \\ -2 & 5 \end{bmatrix} \Rightarrow A^{-1} \cdot B = \frac{1}{4} \begin{bmatrix} 2 & -3 \\ -2 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$

$X = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$

③ $\begin{cases} x - 2y + z = 0 \\ 3x + 2z = 5 \\ 5x - 3y + 3z = 5 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & -2 & 1 \\ 3 & 0 & 2 \\ 5 & -3 & 3 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} 0 \\ 5 \\ 5 \end{bmatrix}}_{B}$

$\Rightarrow \det(A) = 1 \underbrace{\begin{bmatrix} 0 & 2 \\ -3 & 3 \end{bmatrix}}_{6} + 2 \underbrace{\begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}}_{-1} + 1 \underbrace{\begin{bmatrix} 3 & 0 \\ 5 & -3 \end{bmatrix}}_{-9}$

$6 + 2 - 9 = -5 \neq 0$

$A^{-1} = \frac{1}{\det(A)} \times {}^{t}com(A)$

$com(A) = \begin{bmatrix} 6 & +1 & -9 \\ 3 & -2 & -7 \\ -4 & 1 & 6 \end{bmatrix} \Rightarrow {}^{t}com(A) = \begin{bmatrix} 6 & 3 & -4 \\ 1 & -2 & 1 \\ -9 & -7 & 6 \end{bmatrix}$

$A^{-1} \times B = -\frac{1}{5} \begin{bmatrix} 6 & 3 & -4 \\ 1 & -2 & 1 \\ -9 & -7 & 6 \end{bmatrix} \begin{bmatrix} 0 \\ 5 \\ 5 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

4) $\begin{cases} 3x+y+z=6 \\ 2x-3y+2z=-1 \\ x+2y-3z=7 \end{cases}$ $\qquad \underbrace{\begin{bmatrix} 3 & 1 & 1 \\ 2 & -3 & 2 \\ 1 & 2 & -3 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} 6 \\ -1 \\ 7 \end{bmatrix}}_{B}$

$$\det(A) = \underbrace{3\begin{vmatrix} -3 & 2 \\ 2 & -3 \end{vmatrix}}_{15} - \underbrace{1\begin{vmatrix} 2 & 2 \\ 1 & -3 \end{vmatrix}}_{+8} + \underbrace{1\begin{vmatrix} 2 & -3 \\ 1 & 2 \end{vmatrix}}_{+7} = 30 \neq 0$$

$$Com(A) = \begin{bmatrix} 5 & +8 & 7 \\ 5 & -10 & -5 \\ 5 & -4 & -11 \end{bmatrix} \Rightarrow {}^{t}com(A) = \begin{bmatrix} 5 & 5 & 5 \\ 8 & -10 & -4 \\ 7 & -5 & -11 \end{bmatrix}$$

$$\Rightarrow \frac{1}{30}\begin{bmatrix} 5 & 5 & 5 \\ 8 & -10 & -4 \\ 7 & -5 & -11 \end{bmatrix}\begin{bmatrix} 6 \\ -1 \\ 7 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

Exo2:

1) $\begin{cases} x+4y=6 \\ 3x-2y=4 \end{cases}$ $\qquad \begin{bmatrix} 1 & 4 \\ 3 & -2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$

$x = \frac{Dx}{D}$; $D = \det(A) = -14 \neq 0$

$Dx = \begin{bmatrix} 6 & 4 \\ 4 & -2 \end{bmatrix} = -28 \Rightarrow x = \frac{Dx}{D} = \frac{-28}{-14} = \boxed{2}$

$Dy = \frac{Dy}{D} \Rightarrow Dy = \begin{bmatrix} 1 & 6 \\ 3 & 4 \end{bmatrix} = -14$

$y = \frac{-14}{-14} = \boxed{1}$

2) $\begin{cases} 6x+3y=9 \\ 3x+2y=5 \end{cases}$ $\qquad \begin{bmatrix} 6 & 3 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \end{bmatrix}$

$\Rightarrow x = \frac{Dx}{D}$; $D = \det(A) = 3 \neq 0$

$Dx = \begin{bmatrix} 9 & 3 \\ 5 & 2 \end{bmatrix} = 3 \Rightarrow x = \boxed{1}$

$Dy = \begin{bmatrix} 6 & 9 \\ 3 & 5 \end{bmatrix} = 3 \Rightarrow y = \boxed{1}$

TD3 (suite)

Exo2 (suite)

$$\begin{cases} x+y-z=2 \\ 3x+y+2z=9 \\ -x+5y+2z=5 \end{cases} \Rightarrow \begin{bmatrix} 1 & 1 & -1 \\ 3 & 1 & 2 \\ -1 & 5 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 9 \\ 5 \end{bmatrix}$$

$$\Rightarrow x = \frac{Dx}{D} \Rightarrow D = \det(A) = 1\underbrace{\begin{vmatrix} 1 & 2 \\ 5 & 2 \end{vmatrix}}_{-8} - 1\underbrace{\begin{vmatrix} 3 & 2 \\ -1 & 2 \end{vmatrix}}_{-8} - 1\underbrace{\begin{vmatrix} 3 & 1 \\ -1 & 5 \end{vmatrix}}_{-16}$$

$$\Rightarrow -32 \neq 0$$

$$\Rightarrow Dx = \begin{bmatrix} 2 & 1 & -1 \\ 9 & 1 & 2 \\ 5 & 5 & 2 \end{bmatrix} \Rightarrow 2\underbrace{\begin{vmatrix} 1 & 2 \\ 5 & 2 \end{vmatrix}}_{-16} - 1\underbrace{\begin{vmatrix} 9 & 2 \\ 5 & 2 \end{vmatrix}}_{-8} - 1\underbrace{\begin{vmatrix} 9 & 1 \\ 5 & 5 \end{vmatrix}}_{-40}$$

$$\Rightarrow -64 \Rightarrow x = \frac{-64}{-32} = \boxed{2}$$

$$\Rightarrow Dy = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 9 & 2 \\ -1 & 5 & 2 \end{bmatrix} \Rightarrow 1\underbrace{\begin{vmatrix} 9 & 2 \\ 5 & 2 \end{vmatrix}}_{8} - 2\underbrace{\begin{vmatrix} 3 & 2 \\ -1 & 2 \end{vmatrix}}_{-16} - 1\underbrace{\begin{vmatrix} 3 & 9 \\ -1 & 5 \end{vmatrix}}_{-24}$$

$$\Rightarrow -32 \Rightarrow y = \frac{-32}{-32} = \boxed{1}$$

$$Dz = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 9 \\ -1 & 5 & 5 \end{bmatrix} \Rightarrow 1\underbrace{\begin{vmatrix} 1 & 9 \\ 5 & 5 \end{vmatrix}}_{-40} - 1\underbrace{\begin{vmatrix} 3 & 9 \\ -1 & 5 \end{vmatrix}}_{-24} + 2\underbrace{\begin{vmatrix} 3 & 1 \\ -1 & 5 \end{vmatrix}}_{32}$$

$$\Rightarrow -32 \Rightarrow z = \frac{-32}{-32} = \boxed{1}$$

Exo3:

2) $$\begin{cases} 2x+4y=6 \\ 5x+3y=1 \end{cases} \quad \underbrace{\begin{bmatrix} 2 & 4 \\ 5 & 3 \end{bmatrix}}_{A} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \end{bmatrix}$$

$$\det(A) = -14 \neq 0 \Rightarrow \left[\begin{array}{cc|c} 2 & 4 & 6 \\ 5 & 3 & 1 \end{array}\right] \xrightarrow{L_1 \to \frac{1}{2}L_1} \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 5 & 3 & 1 \end{array}\right]$$

$$\xrightarrow{L_2 \to L_2 - 5L_1} \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & -7 & -14 \end{array}\right] \xrightarrow{L_2 \to -\frac{1}{7}L_2} \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 1 & 2 \end{array}\right] \xrightarrow{L_1 - 2L_2} \left[\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & 1 & 2 \end{array}\right]$$

4) $\begin{cases} x + y - 3z = 6 \\ 2x + 3y + 7z = 0 \\ 3x - 2y + 2z = 2 \end{cases}$

$$\underbrace{\begin{bmatrix} 1 & 1 & -3 \\ 2 & 3 & 7 \\ 3 & -2 & 2 \end{bmatrix}}_{A} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 2 \end{bmatrix}$$

$$\det(A) = 1 \underbrace{\begin{vmatrix} 3 & 7 \\ -2 & 2 \end{vmatrix}}_{20} - 2 \underbrace{\begin{vmatrix} 1 & -3 \\ -2 & 2 \end{vmatrix}}_{8} + 3 \underbrace{\begin{vmatrix} 1 & -3 \\ 3 & 7 \end{vmatrix}}_{48} = 76$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -3 & 6 \\ 2 & 3 & 7 & 0 \\ 3 & -2 & 2 & 2 \end{array}\right] \xrightarrow[L_3 - 3L_1]{L_2 - 2L_1} \left[\begin{array}{ccc|c} 1 & 1 & -3 & 6 \\ 0 & 1 & 13 & -12 \\ 0 & -5 & 11 & -16 \end{array}\right] \xrightarrow[L_3 + 5L_2]{L_1 - L_2}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -16 & 18 \\ 0 & 1 & 13 & -12 \\ 0 & 0 & 76 & -76 \end{array}\right] \xrightarrow{L_3 \to \frac{1}{76} L_3} \left[\begin{array}{ccc|c} 1 & 0 & -16 & 18 \\ 0 & 1 & 13 & -12 \\ 0 & 0 & 1 & -1 \end{array}\right] \xrightarrow[L_1 + 16L_3]{L_2 + 13L_3}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array}\right]$$

Exo4:

1) $\begin{cases} x + 3y - 2z + t = 4 \\ 3x - y - t = 0 \\ x + 2y - 3z + 4t = 8 \\ 5x - y + 3t = 10 \end{cases}$

$$\underbrace{\begin{bmatrix} 1 & 3 & -2 & 1 \\ 3 & -1 & 0 & -1 \\ 1 & 2 & -3 & 4 \\ 5 & -1 & 0 & 3 \end{bmatrix}}_{A} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 8 \\ 10 \end{bmatrix}$$

$$x = \frac{D_x}{D} \Rightarrow \det(A) = \left|\begin{array}{cccc|cc} 1 & 3 & -2 & 1 & 1 & 3 \\ 3 & -1 & 0 & -1 & 3 & -1 \\ 1 & 2 & -3 & 4 & 1 & 2 \\ 5 & -1 & 0 & 3 & 5 & -1 \end{array}\right|$$

$$= (1 \times (-1) \times (-3) \times 3) + 0 + (-2 \times (-1) \times 1 \times (-1)) - (\underbrace{(3 \times 3 \times 4 \times 0)}_{0} + (1 \times (-1) \times 3 \times (-1)) + 0)$$

$$\Rightarrow 9 + 2 - 3 = 4$$

$$D_x = \left|\begin{array}{cccc|cc} 4 & 3 & -2 & 1 & 4 & 3 \\ 0 & -1 & 0 & -1 & 0 & -1 \\ 8 & 2 & -3 & 4 & 8 & 2 \\ 10 & -1 & 0 & 3 & 10 & -1 \end{array}\right| = \underbrace{(4 \times (-1)(-3)(3))}_{36} + \underbrace{(-2(-1)(8)(-1))}_{-16} - \underbrace{((4 \times (-1)(-3)(-1}_{12}$$

$$= 36 + -16 + 12$$

$$x = \frac{32}{4} = \boxed{8}$$

Ex01:

$E_3 = \{(x,y,z,t) \in \mathbb{R}^4, x = y = 2z = 4t\}$

- $(0,0,0,0) \in E_3$ car
  $0 = 0 = 2\,0 = 4 \cdot 0$

- $u = (x_1,y_1,z_1,t_1) \in E_3$, $v = (x_2,y_2,z_2,t_2)$

  $u+v = (x_1+x_2, y_1+y_2, z_1+z_2, t_1+t_2)$

  $u \in E_3$, $x_1 = y_1 = 2z_1 = 4t_1$

  $v \in E_3$, $x_2 = y_2 = 2z_2 = 4t_2$

  $x_1+x_2 = y_1+y_2 = 2z_1+2z_2 = 4t_1+4t_2$

  $x_1+x_2 = y_1+y_2 = 2(z_1+z_2) = 4(t_1+t_2)$

  donc $u+v \in E_3$

- Soit $u = (x,y,z,t) \in E_3$, $\lambda \in \mathbb{R}$

  $\lambda u = (\lambda x, \lambda y, \lambda z, \lambda t)$

  $u \in E_3$, $x = y = 2z = 4t$

  $\lambda x = \lambda y = \lambda 2z = \lambda 4t$

  $\lambda x = \lambda y = 2\lambda z = 4\lambda t$

  donc $\lambda u \in E_3$

D'où E_3 est un sous-espace vectoriel.

$E_4 = \{(x,y) \in \mathbb{R}^2, xy = 0\}$

- $(0,0) \in E_4$, car $0 \times 0 = 0$

$u = (1,0) \in E_4$, car $1 \times 0 = 0$

$v = (0,1) \in E_4$, car $0 \times 1 = 0$

$u+v = (1,1) \notin E_4$, car $1 \times 1 = 1 \neq 0$

Donc E_4 n'est pas un sous-espace vectoriel.

$E_5 = \{($

Ex01:

$E_1 = \{(x,y,z,t) \in \mathbb{R}^4 \; ; \; x-y=t \; ; \; x-t=z\}$

$(0,0,0,0) \in E_1$ car

$0-0=0$ et $0-0=0$

$U = (x_1, y_1, z_1, t_1) \in E_1$

$V = (x_2, y_2, z_2, t_2) \in E_1$

$U+V = (x_1+x_2, y_1+y_2, z_1+z_2, t_1+t_2)$

$U \in E_1$, $x_1 - y_1 = t_1$; $x_1 - t_1 = z_1$

$V \in E_1$, $x_2 - y_2 = t_2$, $x_2 - t_2 = z_2$

$x_1+x_2-(y_1+y_2) = x_1-y_1+x_2-y_2$
$= t_1 + t_2$

$x_1+x_2-(t_1+t_2) = x_1-t_1+x_2-t_2$
$= z_1 + z_2$

donc $(U+V) \in E_1$

$U = (x,y,z,t) \in E_1$, $\lambda \in \mathbb{R}$

$\lambda U = (\lambda x, \lambda y, \lambda z, \lambda t)$

$x-y=t$, $x-t=z$

$\lambda x - \lambda y = \lambda t$, $\lambda x - \lambda t = \lambda z$

$\lambda U \in E_1$

donc E_1 est s.e.v

Ex 2 : P_0

$F = \{P \in E,\ P(0) = P'(0) = 0\}$

. $P \in F$, car $P(0) = P'(0) = 0$

. Soient $P_1, P_2 \in F$

$P_1 \in F,\ P_1(0) = P_1'(0) = 0$

$P_2 \in F,\ P_2(0) = P_2'(0) = 0$

$(P_1+P_2)(0) = P_1(0) + P_2(0) = 0$

$(P_1+P_2)'(0) = P_1'(0) + P_2'(0) = 0$

$(P_1+P_2)(0) = (P_1+P_2)'(0) = 0$, donc $P_1+P_2 \in F$

• Soit $P \in F$, $\lambda \in \mathbb{R}$

$(\lambda P)(0) = \lambda P(0) = \lambda . 0 = 0$, $\lambda P \in F$

F est un s-ev.

P

$x_1 + x_2 - (y_1 + y_2) = x_1 -$

$= t$

$x_1 + x_2 - (t_1 + t_2) = x$

$=$

donc $(U+V) \in$

$U = (x, y, z, t)$

• $\lambda \in \mathbb{R}$

$\lambda U = (\lambda x, \lambda$

$x - y = t$

$\lambda x - \lambda y = \lambda t$

$\lambda U \in E_6$

donc E

Exo3

- $W = \lambda_1 u + \lambda_2 V$

$u = (1,1)$, $V = (1,2)$, $W(1,3)$

$$\lambda_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\begin{cases} \lambda_1 + \lambda_2 = 1 \quad (1) \\ \lambda_1 + 2\lambda_2 = 3 \quad (2) \end{cases}$$

$(2) - (1) \Rightarrow \boxed{\lambda_2 = 2}$

(1) $\lambda_1 + 2 = 1$

$\boxed{\lambda_1 = -1}$

Exo 5 :

$v_1(1\ 1\ 1)$, $v_2(0\ 1\ 1)$; $v(0\ 0\ 1)$

Comme la dimention de la famille $\{v_1, v_2, v_3\} = 3$ = dim $\mathbb{R}^3$ donc pour montrer que v_1, v_2, v_3 est une base il suffit de montrer qu'elle est libre

$\lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3 = 0$

$\lambda_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \lambda_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

$\begin{cases} \lambda_1 = 0 \\ \lambda_1 + \lambda_2 = 0 \\ \lambda_1 + \lambda_2 + \lambda_3 = 0 \end{cases} \iff \begin{cases} \lambda_1 = 0 \\ \lambda_2 = 0 \\ \lambda_3 = 0 \end{cases}$

donc la famille est libre alors elle forme une base

$\lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3 \stackrel{?}{=} v$

$\begin{cases} \lambda_1 = 3 \\ \lambda_1 + \lambda_2 = 1 \\ \lambda_1 + \lambda_2 + \lambda_3 = -4 \end{cases}$

$\begin{cases} \lambda_1 = 3 \\ \lambda_2 = -2 \\ \lambda_3 = -5 \end{cases}$

don le condomé de $V = (3, -2, -5)$

Matrice Exam (2020-2021)

Exo 1:

$$\begin{bmatrix} \cdot & 3 & 2 \\ \cdot & 5 & 2 \\ \cdot & -2 & 1 \end{bmatrix} \Rightarrow A^2 = \begin{bmatrix} 2 & 3 & 2 \\ 1 & 5 & 2 \\ 3 & -2 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 3 & 2 \\ 1 & 5 & 2 \\ 3 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 4+3+6 & 6+15+4 & 4+6+2 \\ 2+5+6 & 3+25-4 & 2+10+2 \\ 6-2+3 & 9-10-2 & 6-4+1 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 13 & 17 & 12 \\ 13 & 24 & 14 \\ 7 & -3 & 3 \end{bmatrix}$$

$$B = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \Rightarrow B^2 = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 9 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 9 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = B^4 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 81 & 0 & 0 & 0 \\ 0 & 0 & 16 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

2) • $\det(A) = 2\begin{vmatrix} 5 & 2 \\ -2 & 1 \end{vmatrix} - 3\begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} + 2\begin{vmatrix} 1 & 5 \\ 3 & -2 \end{vmatrix}$

$\Rightarrow \det(A) = 18 + 18 - 34 = 2 \neq 0$

• $\det(B) = 0$. car le produit de Matrice triangulaire égale 0

• $\det(A^2) = 13\begin{vmatrix} 24 & 14 \\ -3 & 3 \end{vmatrix} - 17\begin{vmatrix} 13 & 14 \\ 7 & 3 \end{vmatrix} + 12\begin{vmatrix} 13 & 24 \\ 7 & -3 \end{vmatrix}$

$\Rightarrow 1482 + 1003 - 2484 = 1$

3)

$$Com(A) = \begin{bmatrix} \begin{vmatrix} 5 & 2 \\ -2 & 1 \end{vmatrix} & -\begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 5 \\ 3 & -2 \end{vmatrix} \\ -\begin{vmatrix} 3 & 2 \\ -2 & 1 \end{vmatrix} & \begin{vmatrix} 2 & 2 \\ 3 & 1 \end{vmatrix} & -\begin{vmatrix} 2 & 3 \\ 3 & -2 \end{vmatrix} \\ \begin{vmatrix} 3 & 2 \\ 5 & 2 \end{vmatrix} & -\begin{vmatrix} 2 & 2 \\ 1 & 2 \end{vmatrix} & \begin{vmatrix} 2 & 3 \\ 1 & 5 \end{vmatrix} \end{bmatrix} \Rightarrow Com(A) = \begin{bmatrix} 9 & 6 & -17 \\ -7 & -4 & +13 \\ -4 & -2 & 7 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \times Com(A) = \frac{1}{2}\begin{bmatrix} 9 & 6 & -17 \\ -7 & -4 & 13 \\ -4 & -2 & 7 \end{bmatrix}$$

Exo2 :

$$M = \begin{bmatrix} 3 & 0 & 1 \\ 2 & 1 & 3 \\ -1 & 1 & 0 \end{bmatrix} \Rightarrow P_3(\lambda) = \det(M - \lambda I_3) = \left| \begin{bmatrix} 3 & 0 & 1 \\ 2 & 1 & 3 \\ -1 & 1 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right|$$

$$\Rightarrow \left| \begin{bmatrix} 3-\lambda & 0 & 1 \\ 2 & 1-\lambda & 3 \\ -1 & 1 & -\lambda \end{bmatrix} \right| \Rightarrow (3-\lambda) \begin{vmatrix} 1-\lambda & 3 \\ 1 & -\lambda \end{vmatrix} + 1 \begin{vmatrix} 2 & 1-\lambda \\ -1 & 1 \end{vmatrix}$$

$$\Rightarrow (3-\lambda)\left((1-\lambda)(-\lambda) - 3\right) + 2 + 1 - \lambda$$

$$\Rightarrow (3-\lambda)\left((1-\lambda)(-\lambda) - 3\right) + 3 - \lambda \Rightarrow (3-\lambda)(\lambda^2 - \lambda) + 3 - \lambda$$

$$\Rightarrow (3-\lambda)(\lambda^2 - \lambda + 1) \Rightarrow P_3(\lambda) = 0 \Rightarrow 3 - \lambda = 0 \text{ ou } \lambda = 3.$$

$$\text{ou } \lambda^2 - \lambda + 1 = 0 \Rightarrow \Delta = 1 - 4 \times 1 \times 1 = -3$$

Les Valeurs Propres : $\lambda_1 = 3$

Exo3 : $(M - \lambda_1 I_3) V_1 = 0 \Rightarrow \left(\begin{bmatrix} 3 & 0 & 1 \\ 2 & 1 & 3 \\ -1 & 1 & 0 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \right) \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

$$\Rightarrow \left(\begin{bmatrix} 0 & 0 & 1 \\ 2 & -2 & 3 \\ -1 & 1 & -3 \end{bmatrix} \right) \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} 0 + 0 + c = 0 \Rightarrow \boxed{c = 0} \\ 2a - 2b + 3c = 0 \Rightarrow 2a - 2b = 0 \\ -a + b - 3c = 0 \Rightarrow -a + b = 0 \end{cases}$$

$$V_1 = \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ a \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

Exo3 :

1) $\begin{cases} 3x - 4y = 10 \\ 5x + 2y = 8 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 3 & -4 \\ 5 & 2 \end{bmatrix}}_{A} \underbrace{\begin{pmatrix} x \\ y \end{pmatrix}}_{X} = \underbrace{\begin{pmatrix} 10 \\ 8 \end{pmatrix}}_{B} \Rightarrow \det(A) = 26 \neq 0$

$$\left(\begin{array}{cc|c} 3 & -4 & 10 \\ 5 & 2 & 8 \end{array}\right) \xrightarrow{L_1 \to \frac{1}{3} L_1} \left(\begin{array}{cc|c} 1 & -4/3 & 10/3 \\ 5 & 2 & 8 \end{array}\right) \xrightarrow[L_2 \to L_2 - 5L_1]{} \left(\begin{array}{cc|c} 1 & -4/3 & 10/3 \\ 0 & 26/3 & -\frac{26}{3} \end{array}\right)$$

$$\xrightarrow[L_2 \to \frac{3}{26} L_2]{} \left(\begin{array}{cc|c} 1 & -4/3 & 10/3 \\ 0 & 1 & -1 \end{array}\right) \xrightarrow[L_1 \to L_1 + \frac{4}{3} L_2]{} \left(\begin{array}{cc|c} 1 & 0 & \boxed{2} \\ 0 & 1 & \boxed{-1} \end{array}\right)$$

Exo 3 (Suite) :

$$\begin{cases} x+3y-z=7 \\ 2x-4y+2z=0 \\ x-2y+2z=-1 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 3 & -1 \\ 2 & -4 & 2 \\ 1 & -2 & 2 \end{bmatrix}}_{A} \underbrace{\begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{X} = \underbrace{\begin{pmatrix} 7 \\ 0 \\ 1 \end{pmatrix}}_{B} \xrightarrow[L_3 \to L_3 - L_1]{L_2 \to L_2 - 2L_1}$$

$$\begin{bmatrix} 1 & 3 & -1 \\ 0 & -10 & 4 \\ 0 & -5 & 3 \end{bmatrix} \Rightarrow \det(A) = 1 \begin{vmatrix} -10 & 4 \\ -5 & 3 \end{vmatrix} = -30 - ((-5)\times 4) = -10 \neq 0$$

$$A X = B \Rightarrow A^{-1} A X = A^{-1} B \Rightarrow X = A^{-1} b$$

$x = \dfrac{D_x}{D}$; $D = \det(A)$

$$D_x = \det \begin{pmatrix} \boxed{7} & 3 & -1 \\ \boxed{0} & -4 & 2 \\ \boxed{1} & -2 & 2 \end{pmatrix} = 7 \begin{vmatrix} -4 & 2 \\ -2 & 2 \end{vmatrix} + 0 + 1 \begin{vmatrix} 3 & -1 \\ -4 & 2 \end{vmatrix}$$

$$= 7 \times (-4) + 2 = -26 \neq 0$$

$$x = \frac{D_x}{D} = \frac{-26}{-10} = \boxed{\frac{13}{5}}$$

$$y = \frac{D_y}{D} \Rightarrow D_y = \det \begin{pmatrix} 1 & \boxed{7} & -1 \\ 2 & \boxed{0} & 2 \\ 1 & \boxed{1} & 2 \end{pmatrix} \Rightarrow -1 \begin{vmatrix} 0 & 2 \\ 1 & 2 \end{vmatrix} - 2 \begin{vmatrix} 7 & -1 \\ 1 & 2 \end{vmatrix} + 1 \begin{vmatrix} 7 & -1 \\ 0 & 2 \end{vmatrix}$$

$$\Rightarrow -2 - 30 + 14 \Rightarrow -18$$

$$\Rightarrow y = \frac{D_y}{D} = \frac{-18}{-10} = \boxed{\frac{9}{5}}$$

$$z = \frac{D_z}{D} \Rightarrow D_z = \det \begin{pmatrix} 1 & 3 & 7 \\ 2 & -4 & 0 \\ 1 & -2 & 1 \end{pmatrix} \Rightarrow 7 \begin{vmatrix} 2 & -4 \\ 1 & -2 \end{vmatrix} + 1 \begin{vmatrix} 1 & 3 \\ 2 & -4 \end{vmatrix}$$

$$\Rightarrow -10 \Rightarrow z = \frac{D_z}{D} = \frac{-10}{-10} = \boxed{1}$$

3)
$$\begin{matrix} L_1 \\ L_2 \\ L_3 \\ L_4 \end{matrix} \begin{cases} x+2y-3z+2t=4 \\ 3x-z-2t=0 \\ -x+4y-3z-t=3 \\ 2x+3z-5t=0 \end{cases} \Rightarrow \begin{bmatrix} 1 & 2 & -3 & 2 \\ 3 & 0 & -1 & -2 \\ -1 & 4 & -3 & -1 \\ 2 & 0 & 3 & -5 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 3 \\ 0 \end{pmatrix}$$

$$\xrightarrow{L_3 \to L_3 - 2L_1} \begin{bmatrix} 1 & 2 & -3 & 2 \\ 3 & 0 & -1 & -2 \\ -3 & 0 & 3 & -5 \\ 2 & 0 & 3 & -5 \end{bmatrix} \Rightarrow -2 \begin{vmatrix} 3 & -1 & -2 \\ -3 & 3 & -5 \\ 2 & 3 & -5 \end{vmatrix} \xrightarrow[L_3 + 3L_1]{L_2 + 3L_1} -2 \begin{vmatrix} 3 & -1 & -2 \\ 6 & 0 & -11 \\ 11 & 0 & -11 \end{vmatrix}$$

$$\Rightarrow +2 \begin{vmatrix} 6 & -11 \\ 11 & -11 \end{vmatrix} \Rightarrow 110 \neq 0$$